

UltraBat: An Interactive 3D Side-Scrolling Game using Ultrasound Levitation

Wei-Hsin Wang
National Taiwan University
Taipei, Taiwan
b08901073@ntu.edu.tw

Hong-En Chen
National Taiwan University
Taipei, Taiwan
b08901058@ntu.edu.tw

Mike Y. Chen
National Taiwan University
Taipei, Taiwan
mikechen@csie.ntu.edu.tw

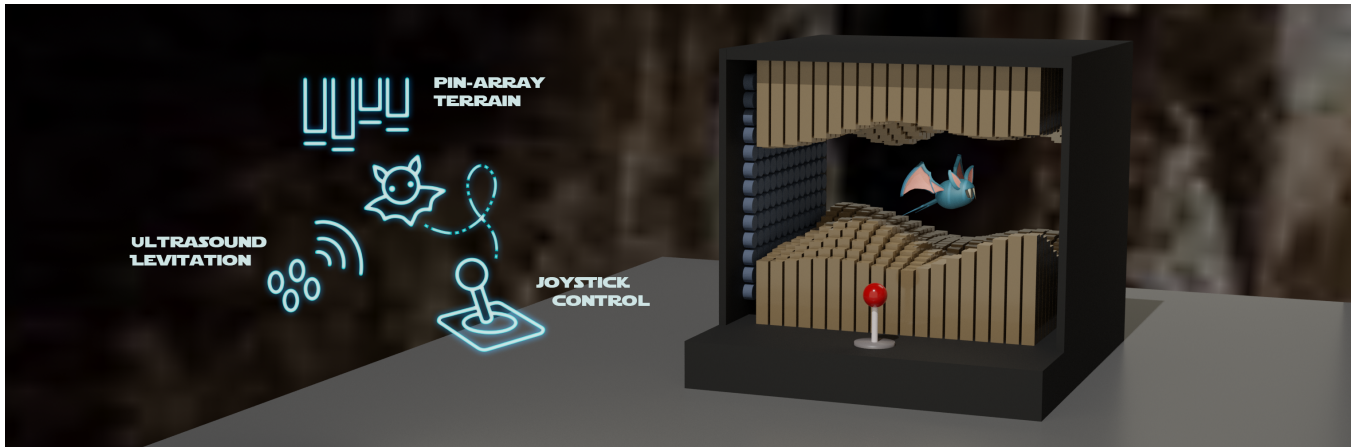


Figure 1: UltraBat side-scrolling game using ultrasound levitation, players control a bat in 3D to avoid hitting moving obstacles.

ABSTRACT

We present UltraBat, an interactive 3D side-scrolling game inspired by Flappy Bird, in which the game character, a bat, is physically levitated in mid-air using ultrasound. Players aim to navigate the bat through a stalagmite tunnel that scrolls to one side as the bat travels, which is implemented using a pin-array display to create a shape-changing passage.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; • **Applied computing** → **Computer games**.

KEYWORDS

arcade game, interactive device, ultrasound levitation, pin arrays

ACM Reference Format:

Wei-Hsin Wang, Hong-En Chen, and Mike Y. Chen. 2022. UltraBat: An Interactive 3D Side-Scrolling Game using Ultrasound Levitation. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3526114.3561344>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3561344>

1 INTRODUCTION

Flappy Bird is a side-scrolling game in which the player controls a bird to fly between columns of green pipes without hitting them. Inspired by this simple yet addictive gameplay, we created a 3D, physical version of this classic called UltraBat. Our system uses ultrasound levitation [3, 4] to provide control of the bat. We use two pin-array displays to create shaping-changing terrains and introduce an ultrasound levitation algorithm to support dynamic obstacles in the control space. Our algorithm minimizes scattering by activating only transducers that has an unobstructed path to the bat. The project source code is available on GitHub¹.

2 GAMEPLAY

UltraBat is a single-player game. The goal of the game is to stay up in the air as long as possible without colliding with the obstacles. The player accumulates points as time goes on. There are LED lights lined on the edges of the box. If the bat gets too close to the stalactites or stalagmites, the box flashes orange light to warn the player. If the bat hits an obstacle, the box will glow red to indicate that the game is over.

We provide two difficulty levels. At standard level, the terrain has a reasonable shape deformation. At crazy level, the terrain updates at a higher speed and forms extreme shape occasionally. Player should beware of random surprise attacks.

¹<https://github.com/ntu-hci-lab/UltraBat>

3 IMPLEMENTATION

UltraBat is played in a box as shown in Figure 1. The pin arrays can move up and down, creating time-varying terrain. Two ultrasound boards levitate the bat between terrains and enable translational movements in 3D space. The joystick controls the motion of the bat. A camera installed behind the box is used to track the position of the bat.

3.1 Terrain: Pin-Based Array

A terrain plane consists of 10×10 3d-printed PLA blocks. Each block is linked to an actuator with a rod-end linkage. A motorized slide potentiometer provides vertical force to the block. The actuators are connected to a customized PCB board and TB6612 motor drivers. With a potentiometer and a DC motor, we can move a block up and down to an exact position. For each block, we can individually control its height according to the map per update. We autogenerated a map of size 10×500 blocks. Every update, each row of blocks shifts its height to match neighboring row. Therefore, the terrain looks like its continuously moving leftward as seen from the view box.

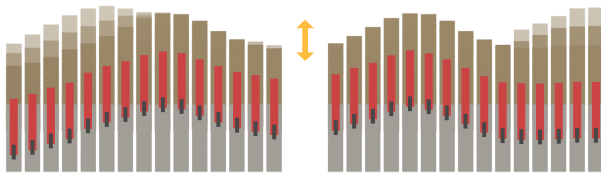


Figure 2: Shape-changing pin arrays: Each block is connected to an actuator that enables vertical movement.

3.2 Ultrasound Levitation

Acoustic levitation manipulates particles in the air by trapping them at regions of standing-wave nodes [2]. By computing phases of the sound field, the algorithm shifts the focus of the trap at a target position. Hence, the position of the bat can be controlled by manipulating ultrasound waves.

In UltraBat, the pin-based terrains obstruct and distort the sound field, so we adopt an algorithm [1] to tackle the presence of arbitrary physical objects. On top of that, we aim to optimize the algorithm furthermore. As the terrain changes shape, some transducers are blocked in the horizontal direction. Our algorithm is programmed with on/off controls for individual transducers to avoid unnecessary reflections caused by hidden transducers. Given the autogenerated terrain map, the height of each block at each update time is known. Therefore, our program precomputes the phase arrays of the sound field for only necessary transducers while switching off the rest.

3.3 Control

UltraBat allows translational movements of the bat, enabling interactive control in the three-dimensional space. The player is provided with arbitrary control of the bat's movement.

A joystick connects to an Arduino Mega that runs our C++ program. Our program translates input commands to ultrasound distributions. Two ultrasound control boards connect the transducers to

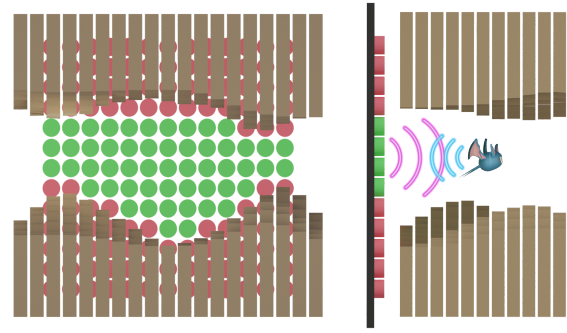


Figure 3: (Left) Transducers are disabled when hidden. (Right) High-speed acoustic algorithm with arbitrary scattering objects.

our microcontroller. With the provided API, Mega forwards data to the control boards to create real-time changes in the sound field. An external adapter provides power to the system. Arduino Mega interfaces with a PC via USB serial communication. The PC is used for manual control and monitoring system. Note that the inclusion of a PC is to ease development. Communication can be done over the Internet to obtain a completely standalone device.

3.4 Extension

There are several extensions to the current design. The terrain is not limited to a monotone color. By replacing some blocks in the top array with an overhead projector, game designers can display various landscapes. The blocks can also be substituted for several different materials, creating visual and audio effects to further improve the game experience and even generate a 4D multisensory presentation. In addition, we also look forward to developing a multi-user game mode in the future. For example, one player is tasked with navigating the bat while the other tries to block its path by manipulating the terrain.

3.5 Potential Adjustments

Implementation depends on the hardware limitations of the provided levitation kit and software flexibility. The arcade box will be designed with respect to the hardware specifications. Also, we will run experiments to determine detailed features, such as an optimal size for the bat, a smooth refresh rate, and a reasonable game difficulty, etc.

REFERENCES

- [1] Ryuji Hirayama, Giorgos Christopoulos, Diego Martinez Plasencia, and Sriram Subramanian. 2022. High-speed acoustic holography with arbitrary scattering objects. *Science Advances* 8, 24 (2022), eabn7614. <https://doi.org/10.1126/sciadv.abn7614>
- [2] Asier Marzo and Bruce W. Drinkwater. 2018. Holographic acoustic tweezers. *Proceedings of the National Academy of Sciences* 116, 1 (2018), 84–89. <https://doi.org/10.1073/pnas.1813047115>
- [3] Rafael Morales, Asier Marzo, Sriram Subramanian, and Diego Martinez. 2019. LeviProps: Animating Levitated Optimized Fabric Structures Using Holographic Acoustic Tweezers. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3332165.3347882>
- [4] Yoichi Ochiai, Takayuki Hoshii, and Jun Rekimoto. 2014. Three-dimensional mid-air acoustic manipulation by ultrasonic phased arrays. *PLoS ONE* 9, 5 (2014). <https://doi.org/10.1371/journal.pone.0097590>